# SadaX – Secure Communication Network

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## About the organization – SadaX.

SadaX is a Saudi startup that aims to provide encrypted messaging and communication solutions and applications. It is specifically targeted at government and private institutions, as well as individuals, particularly those interested in increasing their privacy. The company seeks to create a secure communication environment. It enables users to exchange messages and make calls within a trusted platform. It is also based on high standards of encryption and data governance and adheres to Kingdom of Saudi Arabia regulations on cybersecurity and data protection.

# 1. SadaX Strategic Framework

## 1.1. Vision

We aim for SadaX to be the leading company in the field of secure digital communication within the Kingdom of Saudi Arabia and the Gulf. This will be achieved by providing reliable technological solutions to enhance the privacy of individuals and also for protecting their data from the digital threats and contribute to building a safer cyber environment.

## 1.2. Mission

SadaX's mission is to provide an integrated digital platform that enables users to communicate securely and confidentially. This is achieved by incorporating the latest encryption technologies first, followed by implementing effective data governance policies that ensure data protection from unauthorized access and adhere to local and international regulations related to data protection and cybersecurity, to enhance user confidence.

## 1.3. Strategic Directions

### 1.3.1. Strategy 1: Enhancing End-to-End Encryption

Advanced protection may be a key column of SadaX's vision. In this manner, the company has embraced a clear methodology based on upgrading end-to-end encryption (E2EE) as a essential implication of ensuring client information amid transmission.

General Concept:
End-to-end encryption guarantees that information is scrambled from the sender's gadget and decoded as if it were on the recipient's gadget, without passing through any arrange where the benefit supplier or any mediator party can see the message substance.
Hence, indeed within the occasion of a breach at the server or arrange level, the information remains incoherent and unanalyzed without the private keys held as it were by the clients themselves.

How SadaX Implements This Strategy:

- Adopting Trusted Global Protocols:
  SadaX uses open-source encryption conventions such as the Flag Convention, which is utilized in well-known applications like WhatsApp and Flag, due to its quality, straightforwardness, and fruitful security track record.

- Use of dynamic encryption keys:
  Special encryption keys are produced for each chat or call session, making it troublesome for aggressors to capture or reuse the information in case they get it.

- Full data encryption in transit:
  Uncommon encryption keys are created for each chat or call session, making it troublesome for aggressors to capture or reuse the data in case they get it.

- No key storage on servers:

  Keys are created as it were on users' gadgets, guaranteeing that indeed SadaX's administration cannot get to the substance of the communication. This upgrades client believe and decreases the hazard of information spillage through human intercession or fumble.

- Ongoing technical support:
  SadaX's cybersecurity groups routinely audit the encryption framework and upgrade calculations and conventions in line with worldwide best hones and specialized improvements, such as the progressive move to post-quantum cryptography within the close future.

Benefit achieved:

1. Ensuring client security is genuinely impervious, indeed by company administration.
2. Enhance unwavering quality for person clients and government or private substances concerned with tall privacy.
3. Accomplish compliance with administrative prerequisites such as the PDPL and NCA, which emphasize information assurance amid transmission.

### 1.3.2. Strategy 2: Building a Comprehensive Data Governance Framework

In a quickly developing innovation environment like SadaX, information administration is an vital vital instrument for guaranteeing the secure, organized, and productive dealing

with client information. Subsequently, SadaX receives an progressed methodology pointed at setting up a comprehensive and coordinates information administration system that underpins inside operations, guarantees administrative compliance, and improves client belief.

General Concept:
Data governance is a system of policies and procedures that regulate how data is collected, stored, used, and shared within an organization. It clarifies roles and responsibilities and establishes controls over who can access data, why, and under what circumstances.

Components of the governance framework adopted by SadaX: Data Classification:

All data dealt with inside SadaX is classified into diverse levels of affectability, such as "Exceedingly Private," "Private," "Inside Utilize," and "Open." This classification decides how each sort of information is taken care of and makes a difference avoid unauthorized get to.

Access Control Policies:

Who has the authority to read, modify, or delete data based on their parts and obligations. Role-Based Get to Control (RBAC) frameworks are utilized to absolutely uphold these approaches. Definition of Roles and Responsibilities:

Access Control Policies:

Who has the authority to read, modify, or delete data based on their roles and responsibilities. Role-Based Access Control (RBAC) systems are used to precisely enforce these policies.

Definition of Roles and Responsibilities:

The framework includes clear definitions of responsibilities, such as:

Data Owner: Determines how data is handled.

Data Steward: Monitors data quality and compliance with policies.
Information Security Officer (CISO): Monitors the implementation of security controls and ensures compliance.

Data Quality Controls:
Incorporates intermittent reviews to verify the exactness, completeness, and consistency of data, and to remove copy or adulterated information.

Employee Governance Training:

Mindfulness sessions are conducted for all workers to familiarize them with administration arrangements and how to apply them in their day-by-day work.

Integration with Compliance Systems:

The governance framework is specifically connected to regulatory compliance measures such as the PDPL, making administration a compelling instrument not as it were for data security but also for legal compliance.

Benefit achieved:

- Decreased potential for inner information abuse.
- Made strides operational productivity through clarity of parts and approaches.
- Given clear direction for any inner or outside review.
- Upgraded company validity with controllers and clients.

### 1.3.3. Strategy 3: Achieving Regulatory Compliance

In light of the complex cyber environment and the progressing improvements in regulations and legislation, regulatory compliance is one of the columns of SadaX's success and maintainability as an encrypted communications company. In this manner, SadaX has developed a clear strategy pointed at accomplishing full compliance with national and universal controls and frameworks related to information security and cybersecurity.

General Concept:
Regulatory compliance is the company's commitment to implementing policies and strategies forced by regulatory authorities to ensure that inner operations are conducted in agreement with endorsed legal standards, whether in terms of data capacity, privacy protection, or chance management.

Regulatory Authorities SadaX Complies with:
Saudi Personal Data Protection Law (PDPL):
SadaX is committed to implementing the requirements contained in this law, such as:

- Obtaining explicit user consent before collecting their data.
- Not transferring data outside the Kingdom except under specific controls.
- Enabling the user to access, update, or request deletion of their data.
- Report any data breach within 72 hours to the relevant authorities.
- National Cybersecurity Authority (NCA):

- The company adheres to the authority's controls, such as the Essential Cybersecurity Controls ECC-1:2018, which include:
- Establishing a Security Operations Center (SOC) to monitor and respond to attacks.
- Activating a SIEM to monitor system behavior and analyze incidents.
- Monitoring and updating endpoints.
- Conducting periodic risk assessments and penetration tests.
- Saudi Arabian Monetary Authority (SAMA): (When handling financial information)
- Implementing digital governance controls specific to the financial sector.
- Enable multi-factor authentication for all sensitive operations.
- Encrypting data in transit and storage using SAMA-approved technologies.
- Aramco Cyber Compliance Certification (ARAMCO CCC): (If SadaX is used in industrial or oil environments)
- Adhering to third-party risk assessment standards.
- Submitting periodic vulnerability reports.
- Implementing a Secure Software Development Lifecycle (SSDLC).
- Penetration testing conducted by an accredited independent entity.

How SadaX implements this strategy:

- Assigning a dedicated team to continuously monitor regulatory changes.
- Archive all compliance-related policies, procedures, and documents.
- Conduct awareness and training sessions for employees on their responsibilities in adhering to these regulations.

Benefit achieved:

- Maintaining a strategic distance from punishments and fines related with noncompliance.
- Upgrading client certainty that their information is being prepared to the most elevated security measures.
- Expanding SadaX's availability to collaborate with government and private divisions with exacting administrative necessities.
- Qualifying the company for formal cybersecurity certifications.

**Building a comprehensive data governance framework**

The company aims to design and implement clear policies for data classification, controlling access, distributing permissions, and conducting periodic reviews, thus contributing to the protection of digital data assets from internal and external threats.

**Achieving regulatory compliance:**

SadaX works to fully comply with applicable regulations in the Kingdom, such as the Personal Data Protection Law (PDPL), the requirements of the National Cybersecurity Authority (NCA), and the Capital Market Authority (SAMA), to ensure the reliability of its services and enhance customer confidence.

## 1.4. Objectives

- Implementing a multi-factor authentication (MFA) system for all users by the end of this year to enhance account security.
- Implementing a comprehensive digital data classification plan based on its sensitivity within the first six months of the official launch.
- Publishing a clear and detailed privacy policy in line with the PDPL regulations, available via the online platform and app.
- Obtaining official accreditation from the National Cybersecurity Authority within the first year of the platform's operation.
- Establishing an internal Security Operations Center (SOC) to monitor cyber activities and anticipate threats, ensuring zero breaches in the digital environment.

# 2. Data Process

## 2.1. Data Classification Process

For SadaX – Encrypted Communications Network

### 2.1.1. Process Definition:

Data classification is a fundamental process for organizing an organization's data according to its sensitivity and protection requirements. This helps determine how it is handled in terms of access, storage, and transmission.

In the context of SadaX, which handles highly sensitive data such as encrypted messages and system logs, data classification is the first step in building an effective governance framework that protects information assets from cyber threats and enhances the system's ability to achieve regulatory compliance and protect privacy.

### 2.1.2. Data Classification Policy:

SadaX is committed to implementing a strict data classification policy, which aims to ensure the secure and orderly handling of all types of data managed by the platform. The policy stipulates:

- Data classification is based on its sensitivity and potential impact if exposed to any threat (leakage, loss, unauthorized modification).
- Access to classified data is restricted based on specific permissions granted to employees based on their roles and responsibilities.
- Conduct a periodic review of classifications every six months, or when changes occur in the nature or use of the data.
- Integrate classification into data management systems so that it is clearly defined for each record or file being handled.

### 2.1.3. Suggested Classification Levels:

| Classification Level | Definition | Examples |
|---|---|---|
| **Highly Confidential** | Data that is extremely sensitive, and any disclosure could result in significant harm to the organization or its users. | Encryption keys, access logs, encryption server data |
| **Confidential** | Data that must be protected from public access; its leakage could cause limited or moderate harm. | User account data, encrypted messages, session records |
| **Internal Use Only** | Data intended for internal use within organizational departments and should not be shared externally. | Performance reports, operational schedules, internal emails |
| **Public** | Data that does not require special protection and can be freely shared without risk. | Service information, user manuals, public announcements |

### 2.1.4. Steps to Implement the Classification Process:

1. Define Data Types: Text messages, calls, registration data, system logs.
2. Sensitivity Level Analysis: Evaluating each type of data based on the wildness of damage that could occur if it is leaked or tampered with.
3. Apply Classification Levels: Data classified according to the four levels described above, based on the results of the analysis.
4. Authority Mapping: determining who has access to each category of data is based on their role and the level of trust they have.
5. Update Classification Periodically: reviewing periodically at least every six months, And if necessary, when there are significant changes in the type or use of data.

### 2.5. Tools used:

- Excel Sheet/Google Sheets: To create a classification tracking table
- Internal Tagging System: Each file or record is automatically assigned a classification
- DLP Tool: To prevent data leakage based on classification

### 2.1.5. Plugin: Procedures for handling sensitive data when handling "Highly Confidential":

Given the risk of leakage of "top secret" data, it is handled according to strict standards, including:

- Prohibiting its transmission via email or any other external means of communication.

- Restricting its use within the SadaX network only, prohibiting its transfer to third parties.
- Permanently encrypting it automatically.
- Documenting all operations performed on this data, including access and modification.

**2.1.6 Additional Component: Data Breach Notification Plugin**

As portion of SadaX's progressing efforts to reinforce data governance and align with national regulations, a Data Breach Notice Plugin has been coordinates into the platform's security infrastructure. This component plays a crucial part in improving transparency, accountability, and fast response within the event of any data compromise.

The plugin works in conjunction with the organization's Security Data and Event Management (SIEM) system. It is consequently activated upon the discovery of unauthorized get to, irregular behavior, or suspected information spillage.

Core Functions:

Consequently, cautions the Information Security Officer and the Security Operations Center (SOC) in genuine time upon discovery of any suspicious action.

Creates organized incident reports that include points of interest such as time, influenced data sorts, included client accounts, and reaction activities taken.

Triggers the platform's occurrence reaction convention, counting data isolation, legal examination, and prompt control measures.

Ensures compliance with the Individual Data Protection Law (PDPL), which needs informing the Saudi Data & AI Authority (SDAIA) inside 72 hours of a affirmed data breach.

This plugin reflects SadaX's commitment to ensuring its clients and their data. It makes a difference the organization to respond quickly and responsibly to incidents, minimize the effect of breaches, and keep up believe and compliance in a highly sensitive digital environment.

## **2.2.** Data Privacy Process – Explanation

Privacy is at the core of SadaX's plans as an encrypted communication solutions provider. The data privacy procedure secures users' data in line with regulatory standards like Saudi Arabia's Personal Data Protection Law (PDPL) to safeguard it in all stages of its life cycle collection, handling, storage, and sharing.

### **2.2.1. Privacy Policy:**

- Personal data will not be collected without the expressed consent of the user.
- Data is exclusively used for purposes declared (e.g., user authentication, registration).
- All data pertaining to individuals has to be encrypted in storage and transit using AES-256 or equivalent.
- Access to the individual's data is strictly monitored and role based.
- Privacy Impact Assessment (PIA) is required prior to the launch of any data-focused feature.
- Retention periods are established according to the type of data and legal requirements.
- Users are entitled to request deletion of or corrections to their data.

#### 2.2.2. Implementation Steps:

1. List all types of personal data processed through the platform.
2. Map data flows to determine where the personal data is being stored or transmitted.
3. Including consent mechanisms and privacy settings options in the app.
4. Implement internal data breach response procedures.
5. Provide employee privacy and awareness training sessions.

#### 2.2.3. Tools Utilized

- Data Masking Software (e.g. Informatica) to anonymize the sensitive personal data.
- DLP (Data Loss Prevention) Systems to impose policies on sensitive data. Consent Management Platforms to monitor and apply user data permissions.
- Encryption Tools (such as OpenSSL) to protect the user data in motion and at rest.

## 4. Data Dictionary

**Table 1: Users**

| Key | Name | Data Type | Null | References | Description |
|---|---|---|---|---|---|
| **PK** | UserID | int | No | - | Unique ID assigned to each user. |
| | FullName | varchar(100) | No | - | Full legal name of the user. |
| | PhoneNumber | varchar(20) | No | - | Registered phone number. |
| | EmailAddress | varchar(100) | No | - | User's email address. |
| | AccountCreationDate | date | No | - | Date and time when the account was created. |
| | MFA_Enabled | BIT | No | - | Whether multi-factor authentication is enabled. |
| | Status | varchar(10) | No | - | Account status (e.g., Active, Suspended). |

**Table 2: Communication Log**

| Key | Name | Data Type | Null | References | Description |
|---|---|---|---|---|---|
| **PK** | MessageID | int | No | - | Unique identifier for each message. |
| **FK** | SenderUserID | int | No | Users | References sender's user ID. |
| **FK** | ReceiverUserID | int | No | Users | References receiver's user ID. |
| | MessageTimestamp | date | No | - | Timestamp of when the message was sent. |
| | MessageLength | Int | Yes | - | Number of characters in the message. |
| | EncryptionType | varchar(20) | No | - | Type of encryption used (e.g., AES-256, RSA). |
| | DeliveryStatus | varchar(20) | Yes | - | Status of the message (e.g., Sent, Delivered, Failed). |

**These tables represent the center database structure of the SadaX stage. The Clients table stores basic account data, whereas the Communication Logs table records encrypted message exchanges between clients. Primary and foreign key constraints are actualized to ensure referential integrity and secure data relationships.**

**The following Entity Relationship Diagram (ERD) illustrates the logical relationship between the two main database tables in the SadaX platform: Users and Communication Logs.**

**The diagram shows how each message is linked to a sender and a receiver through foreign key references to the Users table, ensuring secure, traceable communication between users.**

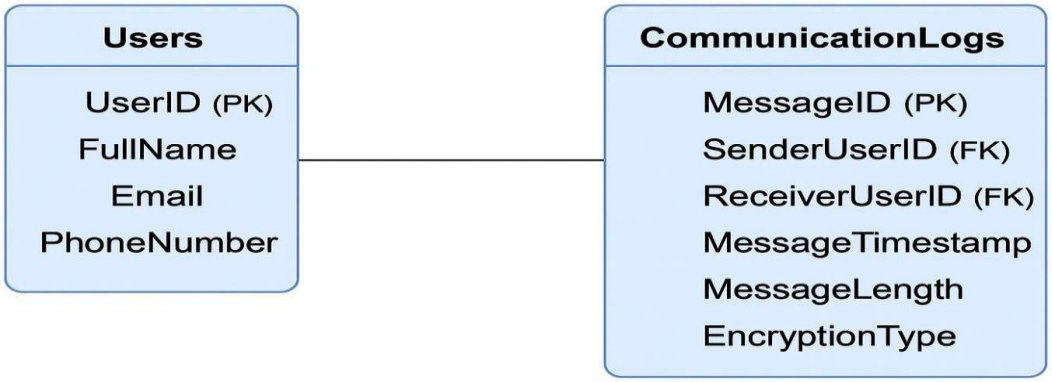


**Figure 1**

**Appendix A – SQL Table Structures**
The SQL code used to create the main database tables for the SadaX platform has been provided in a separate attachment.

This code defines the structure of the Users and Communication Logs tables, including column names, data types, primary and foreign key constraints, and comments on each field.

**The file is titled:**
**Sadax_DataDictionary.sql**

and is included with the submitted materials on Blackboard

## 5. Regulatory & Compliance

SadaX is dedicated to ensuring compliance with local and sector-specific regulatory authorities of the Kingdom of Saudi Arabia. The following is a brief overview of compliance activities:

### 5.1 Personal Data Protection Law (PDPL):

- Explicit consent is a prerequisite to collecting personal data.
- Personal data is hosted solely on servers in Saudi Arabia.
- Data breach notifications are reported to the Saudi Data & AI Authority (SDAIA) within 72 hours.
- Data subjects are entitled to full rights to access and rectify their data.

**5.2 National Cybersecurity Authority (NCA):**

- Adheres to the Essential Cybersecurity Controls (ECC-1:2018).
- Security Operations Center (SOC) constantly monitors and logs all activities.
- Endpoints are all hardened and are monitored using SIEM systems.
- Compliance audits are done annually and documented.

**5.3 Saudi Arabian Monetary Authority (SAMA):**

- Adopts cybersecurity standards as per SAMA's Cybersecurity Framework.
- Adheres to least-privilege practices in terms of financial or sensitive user interactions.
- Data is protected both in transit and in storage with SAMA-approved mechanisms.
- Identity verification involves biometric and multi-factor authentication.

**5.4 ARAMCO Cybersecurity Compliance Certificate (CCC):**

- Verifies platform adherence to ARAMCO's third-party risk assessment standards.
- Compliance documents and reports on vulnerabilities are dispatched to ARAMCO.
- Regular penetration testing is conducted by qualified third-party providers.
- Adheres to secure software development life cycle (SSDLC).

## 6. Data Governance Policy

**6.1 Objective:**

This policy aims to establish a comprehensive framework for managing data within the SadaX network to ensure its confidentiality, integrity, and availability. It enables the organization to protect its encrypted communications from cyber threats and data breaches, while supporting compliance with privacy regulations and both national and international legal frameworks.

**6.2 Scope:**

The Data Governance Policy applies to all data collected, stored, processed, or shared through the SadaX platform, including text messages, calls, user data, and encrypted content. It covers system administrators, end users, and developers involved in the platform's security infrastructure.

### 6.3 Roles and Responsibilities:

**Data Owner:** Responsible for defining data classifications and access rights.

**Data Steward:** Ensures data quality and adherence to governance policies.

**Chief Information Security Officer (CISO)**: Oversees implementation of security controls and compliance.

**End User:** Required to access and use data in accordance with its classification and access policies.

## 7. Data Standards

### 7.1 Types of Data in SadaX:

1. **Personal Data**: Usernames, phone numbers, email addresses, login identifiers.
2. **Professional Data**: Work-related contact details, internal communication records, user roles.
3. **Sensitive Data**: Passwords, encryption keys, biometric authentication data.

### 7.2 Detailed Standards by Data Type:

1. **Structured Data**

Examples: databases with user records, message logs, timestamps.

Standards:

- Consistent field formatting (e.g., dates in YYYY-MM-DD).
- Unique identifiers for each user and communication record.
- Sensitive fields (e.g., passwords) must be clearly labeled and restricted.

2. **Unstructured Data**

Examples: image files, audio recordings, PDF attachments.

Standards:

- All unstructured files must be encrypted during storage and transmission (AES256).

- Only specific file types are permitted (mp3, .pdf, .png, etc.).
- Files must be organized in secure, metadata-based folders for traceability.

3. **Sensitive Data**

Examples: encryption keys, passwords, personal biometric data.

Standards:

- Enforced encryption using AES-256 or RSA with minimum 2048-bit keys.
- Implementation of strict least privilege access control policies.
- All sensitive data must be stored in isolated, secure containers (e.g., virtualized environments).

## 8. Technical Tools

### 8.1 Currently Used:

Manual Tracking: Data classification and access logs are currently maintained through Excel and PDF documentation.

### 8.2 Proposed Tools for Implementation:

1. **Encryption Algorithms**:

AES and RSA to secure both static and transmitted data.

2. **VPN Technology**:

VPN servers (e.g., OpenVPN or WireGuard) to secure user communication channels.

3. **Biometric Authentication**:

Fingerprint and facial recognition for secure user verification.
4.**Access Management Systems**:

Keycloak or Microsoft Entra ID for role-based access control and Single Sign-On (SSO).

5.**Data Cataloging**:

Apache Atlas to document data assets, classification, lineage, ownership, and usage tracking.

**9. Data Inventory**

In a highly sensitive environment like SadaX, which deals with encrypted communications, maintaining a well-structured data inventory is essential. It helps identify data types, storage locations, purposes, and responsible roles. This inventory plays a crucial role in ensuring compliance with Saudi regulations such as PDPL and NCA, and supports the implementation of access controls, encryption, and privacy policies.

| Data Type | Source | Storage Location | Purpose | Responsible Role |
|---|---|---|---|---|
| **Encrypted user messages** | SadaX application | Encrypted internal servers | Secure communication between users | Security development team |
| **Encryption keys** | Internal encryption system | Isolated virtual environments | Encrypting and decrypting user data | Chief Information Security Officer (CISO) |
| **Personal user data** | Registration process | PostgreSQL database | Identity verification and personalized services | Data Privacy Officer |
| **System logs** | Backend system | SIEM platform | Monitoring system behavior and detecting anomalies | SOC |
| **Media files (attachments)** | User input | Encrypted cloud storage | Sending and receiving in-app attachments | Data management team |
| **Privacy settings** | User dashboard | Preferences database | User control over privacy and security configurations | UX and Privacy Team |

## 10. Risk Analysis (FMEA)

Risk analysis through the Failure Mode and Effect Analysis (FMEA) method allows SadaX to proactively identify potential vulnerabilities within its data privacy process. By evaluating severity (S), likelihood of occurrence (O), and ease of detection (D), the Risk Priority Number (RPN) helps prioritize which issues to address first. This method enhances the platform's resilience and user trust.

| Process | Failure Mode | Cause | Effect | S | O | D | RPN | Control Measures |
|---|---|---|---|---|---|---|---|---|
| **Data Encryption** | Encryption key leakage | Inadequate access control | Loss of message confidentiality, system compromise | 9 | 4 | 3 | 108 | Apply MFA, review access, isolate keys |
| | Weak encryption algorithms | Legacy encryption in use | Easier to break encryption | 7 | 3 | 4 | 84 | Upgrade to AES256, disable weak protocols |
| **Access Control** | Unauthorized access | No multi-factor authentication | User privacy violations, legal risk | 8 | 5 | 4 | 160 | Use biometric + OTP, strong authentication |
| **Data Classification** | No clear data classification | Lack of awareness/systems | Sensitive data mishandling | 6 | 4 | 3 | 72 | Auto-tagging, staff training |
| **Data Transmission** | Data via insecure channels | Email or external sharing | Interception during transfer | 8 | 3 | 2 | 48 | Disable external send, enforce internal encryption |

## 11. Data Glossary

The data glossary serves to clarify technical and governance-related terms used throughout the SadaX data framework. This ensures consistent understanding between developers, security teams, and policymakers, especially when dealing with sensitive data and regulatory compliance.

| Term | Definition | Usage in SadaX |
|---|---|---|
| **End-to-End Encryption** | Encryption where only sender and receiver can access the message. | Used to protect communication from third-party interception. |

| | | |
|---|---|---|
| **MFA** | Multi-Factor Authentication; uses more than one method to verify identity. | Required for login and privileged operations. |
| **AES-256** | A 256-bit symmetric encryption algorithm. | Secures stored and transmitted data. |
| **RSA** | Asymmetric encryption using a public/private key pair. | Used in secure key exchange for encrypted sessions. |
| **SIEM** | Security Information and Event Management platform. | Monitors system activity and detects anomalies. |
| **SOC** | Security Operations Center. | Monitors logs and incidents in real-time. |
| **CISO** | Chief Information Security Officer. | Oversees implementation of data governance policies and security controls. |
| **DLP** | Data Loss Prevention. | Prevents unauthorized transmission of sensitive data. |
| **Data Classification** | Organizing data based on its sensitivity and importance. | SadaX classifies data into Highly Confidential, Confidential, Internal, and Public. |
| **Data Inventory** | Central list of all data assets and metadata. | Tracks where data is stored, who owns it, and its usage. |
| **RPN** | Risk Priority Number = Severity × Occurrence × Detection. | Helps rank and prioritize risks in FMEA. |
| **FMEA** | Failure Mode and Effect Analysis. | Used in SadaX to assess privacy-related risks. |
| **Data Dictionary** | Collection of metadata (table names, fields, formats). | Documents database structures and field-level definitions. |

| | | |
|---|---|---|
| **Data Glossary** | List of key business and technical terms used in the organization. | Ensures consistency across documentation and implementation. |
| **PDPL** | Saudi Arabia's Personal Data Protection Law. | SadaX adheres to it to regulate user data usage and processing. |
| **Compliance** | Adhering to regulations and internal policies. | SadaX complies with PDPL, NCA, SAMA, and ARAMCO CCC. |

| | | |
|---|---|---|
| **Access Control** | Mechanism to limit who can access or edit data. | Enforced based on user roles. |
| **Metadata** | Data about data, including origin, structure, and type. | Managed through Apache Atlas in SadaX. |
| **Encryption Key** | A piece of information that determines the output of encryption. | Used in encrypting messages and user data. |
| **Privacy Policy** | Rules that dictate how personal data is collected and handled. | Required for transparency and PDPL compliance. |
| **Audit Trail** | Logs that record access and activity related to data. | Maintained in SadaX's SIEM system. |
| **Data Steward** | Role responsible for maintaining data quality and compliance. | Ensures SadaX data governance policies are enforced. |
| **SSDLC** | Secure Software Development Life Cycle. | Followed to ensure SadaX platform is secure by design. |
| **Data Lifecycle** | The stages data goes through collection, use, storage, destruction. | Governed by retention and deletion policies. |
| **User Table** | A database table that stores user-specific information such as ID, email, and authentication data. | Used to manage user accounts, identities, and access within SadaX. |
| **Message Table** | A table used to store encrypted messages and metadata such as timestamps and sender IDs. | Stores end-to-end encrypted communication data for auditing and delivery. |
| **Field Name** | A specific column in a database table representing a data attribute. | Used to identify and retrieve user or message attributes in queries. |
| **Data Type** | Defines the kind of data stored in a field such as string, integer, or timestamp. | Ensures data is stored and processed in valid formats in SadaX. |
| **Description** | An explanation of the purpose or contents of a field or table. | Helps developers and analysts understand database structure and logic. |

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